

Imad Ahmad – MSc. Data Science

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SUMMARY

Production-focused Machine Learning Engineer with deep **ML Development and Production Deployment** experience. Shipped a call-offload model that delivered an **estimated \$1.1M in annual savings** and a custom-loss churn predictor trained on millions of records that provided insights directly to **Director Level Leadership**. Having published academic statistical research, I combine statistical depth with production engineering to turn rigorous modeling into systems that move real business metrics.

EDUCATION

Master of Data Science and Analytics, *University of Calgary*

Graduated December 2023

Bachelor of Science in Integrated Science, *University of British Columbia*

Graduated May 2020

EXPERIENCE

TELUS Communications Inc.

October 2024 – Present

Machine Learning Engineer | July 2025 - Present

- Implemented a proactive call-offload initiative as **lead developer**, working directly with stakeholders to ship a **PyTorch deep learning model** to production via **Kubeflow pipelines on GCP Vertex AI** — delivering **\$1.1M in annual operational savings**.
- Developed and **took ownership** of an ML evaluation service that is used by over **40 teams company wide**, processing **400-700 evaluation requests per month** and providing estimated **annual savings of \$750K**.
- Built a reusable ML observability framework — optimizing drift, data quality, and latency monitoring across pipelines using **Vertex AI Model Monitoring** and **Arize** — adopted so widely it became the team's default instrumentation layer for every new model deployment.
- Identified the statistical drivers of **customer churn** across millions of records by building a **PyTorch model** with a **custom optimization routine** to properly weight rare-but-costly churn events, translating findings into a retention strategy adopted by **director-level leadership**.
- Designed and implemented end-to-end **MLops pipelines on GCP Vertex AI using Kubeflow** - automating data preparation, training, evaluation, and deployment across over **one million customer records** - with integrated alerting and failure detection that **cut issue resolution time by 30%**.

Data Engineer | October 2024 – July 2025

- Spearheaded a cross-functional migration initiative involving multiple engineering teams to execute a **zero-downtime** transition of **100+ mission-critical ETL pipelines** to GCP Cloud Composer 3, meeting SLAs for **99.9% data availability**.
- Architected and led the QA automation strategy for the **Telus-WestJet strategic partnership**, designing **Python-based data validation pipelines** that reconciled cross-company architectures and guaranteed **99.99% accuracy** in high-volume customer identity synchronization.
- Designed a **modular DAG generation framework** within Cloud Composer using **Python** and custom operators, allowing teams to deploy configuration-based pipelines without writing repetitive boilerplate code.

LifeLabs Inc.

August 2021 – July 2022

Laboratory Data Assistant | August 2021 – July 2022

- Designed and deployed 15+ complex **Apache Airflow DAGs** to automate **critical ETL pipelines**, integrating **CI/CD workflows** to ensure version-controlled releases and system reliability - reducing overall data processing time by 40%.
- Optimized daily extraction and transformation of **1,000+ COVID-19 test records** through **high-performance SQL queries** and implementing observability via **Prometheus** and **Grafana**, improving operational reliability and decision-making.

PUBLICATIONS

Academic Statistical Research (Published) – Exploring the Gender Disparity in Physiology Departments

<https://pubmed.ncbi.nlm.nih.gov/33409087/>

- Utilized advanced statistical methods in **Python** and **R** to explore the gender disparity in Physiology departments for over 2000 faculty members, contributing valuable insights into diversity dynamics within academic settings.
- Authored and **successfully published** research in a peer-reviewed journal as the **first author**, demonstrating proficiency in scientific communication and scholarly writing.

CERTIFICATIONS

GCP Associate Cloud Engineer • 2026

https://www.credly.com/badges/4f7319a2-c515-40e9-a562-a6f7d8f3d615/public_url

GCP Cloud Digital Leader • 2024

https://www.credly.com/badges/72caaec7-b019-4bc1-a2e0-c7f2ad21b16a/public_url

AWS Certified Cloud Practitioner • 2023

https://www.credly.com/badges/22444caa-2cfa-4a7d-ade2-4608ca8922a0/public_url